QEG-RAG: Query-Time Evidence Graphs with Path-Scaffolded Diagnostic Reasoning for Faithful Table–Text Question Answering

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Abstract. Retrieval-augmented generation (RAG) over heterogeneous databases requires resolving complex queries across semi-structured tables and unstructured text passages. Under strict context-grounded constraints where models must reason exclusively over retrieved evidence without parametric hallucination, answering multi-hop queries demands cross-modal retrieval—anchoring candidate records in structured tables and traversing external links to retrieve supporting text passages. However, existing paradigms face a structural dilemma: standard RAG linearizes tables into flat strings, dissolving horizontal row-level bindings and conflating symmetric records, while recent graph-based methods either discard critical peripheral facts via aggressive pruning or incur high latency through unbounded agent loops. In this work, we propose QEG-RAG, centered on *path-scaffolded diagnostic reasoning over query-time evidence graphs*. At query time, QEG-RAG unifies retrieved tabular and textual evidence into a schema-preserving multigraph using star-topology record centroids to enforce horizontal row bindings. Rather than discarding context via pruning, it preserves complete evidence through compact prompt serialization while extracting deterministic, multi-factor scored reasoning paths to guide language model attention across cross-modal hops. To resolve initial retrieval ambiguity such as candidate collisions or missing passages, QEG-RAG employs a bounded two-tier execution protocol: confident queries conclude in a single primary call, whereas structural ambiguities trigger a diagnostic handoff directing targeted, candidate-balanced bridge retrieval to verify competing candidates factually within at most two calls. Empirical evaluations on multi-hop benchmarks demonstrate that QEG-RAG consistently improves grounding accuracy and reduces ungrounded answers under strict evidence constraints with bounded query-time efficiency.

Keywords: Table question answering · Knowledge graphs · Retrieval-augmented generation · Path scaffolding · Faithful reasoning

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1 Introduction

Modern question answering increasingly operates over *heterogeneous databases* where relational attributes are structured in two-dimensional tables while contextual narratives reside in unstructured text passages [1]. In such hybrid corpora, resolving complex queries requires *cross-modal multi-hop retrieval* [2]: neither modality alone contains the complete answer. For example, in a representative query such as *"What was the original name of the city with an AFL club formed in 1989?"* (Fig. 2), the retrieval pipeline must first anchor candidate records within a sports table matching the attribute `Formed = 1989`, identify the bridge entity `City = Toronto`, and subsequently retrieve external text passages linked to Toronto to uncover its historical name (*York*).

However, standard retrieval-augmented generation (RAG) paradigms face fundamental representation and inference bottlenecks. Conventional retrievers linearize tables into raw token strings, dissolving horizontal row boundaries and column hierarchies, which induces severe *attribute conflation* across symmetric records. To capture cross-modal dependencies, recent paradigms construct knowledge graphs (KGs) directly at query time [3, 4]. Yet, existing online KG methods rely either on expensive LLM-based triple extractors or flat heuristics, and span two problematic extremes: zero-shot destructive graph pruning (e.g., ODYSSEY [3]) and unbounded agent-driven exploration loops over online KGs (e.g., HydraRAG [5]).

A systematic examination of existing hybrid RAG frameworks reveals four fundamental limitations that directly impede faithful multi-hop reasoning:

1. *Absence of explicit reasoning guidance vs. destructive pruning*: Existing methods typically feed the constructed graph into the language model as an unorganized collection of edges, causing attention to disperse across irrelevant entities. Conversely, aggressive pruning (e.g., ODYSSEY) discards critical peripheral facts (qualifying dates, secondary constraints) to fit context limits, forcing models into ungrounded guessing or empty failures.
2. *Retrieval ambiguity, structural collisions, and heuristic tiebreaking*: Initial multimodal retrieval is frequently under-determined: multiple candidate records often satisfy partial query constraints, or critical bridge passage links are missing. Existing single-turn systems lack diagnostic self-audit and resort to blind guessing or heuristic lexical tiebreaking over competing candidates rather than factual verification.
3. *Loss of tabular semantics and costly graph extraction*: When converting tables into graph triples, standard extraction decomposes records into isolated pairwise facts, stripping column headers and row context. Cells detach from originating records, causing horizontal attribute conflation across symmetric rows. Furthermore, relying on LLM prompting for relation extraction incurs prohibitive latency and cost during online query execution.
4. *Operational dilemma: fragility vs. unbounded agent overhead*: Systems face a sharp trade-off: rigid single-turn execution fails irreparably when facing retrieval gaps, whereas agentic online KG exploration frameworks (e.g., HydraRAG, ToG-3) deploy open-ended iterative loops that incur high latency

and dozens of expensive LLM calls per query without bounding worst-case costs.

To address these challenges, we introduce QEG-RAG (query-time evidence graphs for RAG). Rather than relying on costly LLM extraction, unorganized triple pools, or unbounded agent loops, QEG-RAG preserves semantic relational bindings and establishes *explicit reasoning guidance over online multigraphs coupled with diagnostic-driven retrieval feedback*. Specifically: (1) *Explicit path-scaffolded reasoning*: Rather than discarding context via destructive pruning, the framework retains complete evidence through compact prompt serialization while extracting deterministic, multi-factor scored reasoning paths p_1 ($S(p)$) that steer language model attention along verified cross-modal relations as non-filtering reading scaffolds. (2) *Diagnostic-driven retrieval feedback*: When structural ambiguity arises, the model defers ungrounded generation and emits a structured diagnostic packet triggering targeted, candidate-balanced bridge retrieval; the gathered facts are packaged into a minimal sufficient state capsule $\mathcal{K}_2$ for factual resolution in a bounded second call. (3) *LLM-free star multigraph construction*: QEG-RAG deterministically instantiates an online multigraph per query in $< 20\,\text{ms}$ (0 LLM calls) via star-topology record centroids ρ_i. These centroids explicitly bind cell values to normalized column predicates, mathematically preventing cross-row attribute conflation. (4) *Bounded two-tier execution protocol*: Unambiguous queries conclude immediately in a single primary call (fast exit), while ambiguous cases are resolved within at most a targeted second call, strictly bounding execution overhead.

Empirical evaluations across multi-hop table–text benchmarks (HybridQA and OTT-QA) demonstrate that QEG-RAG consistently improves grounding accuracy and reduces ungrounded answers over direct prompting, flat retrieval, and graph baselines, while maintaining bounded query-time efficiency.

We articulate four principal contributions directly resolving these challenges:

- *Explicit reasoning guidance over online multigraphs*: A paradigm that reconciles query-time, schema-preserving multigraph construction with deterministic multi-factor path scoring ($S(p)$), steering language model attention along verified cross-modal relations without destructive evidence.
- *Diagnostic-driven retrieval feedback*: A controlled handoff mechanism where structural collisions or missing links trigger targeted, candidate-balanced bridge retrieval, resolving bottlenecks factually rather than through heuristic tiebreaking.
- *Preservation of tabular semantics*: A star-topology centroid formulation that explicitly maintains horizontal row bindings and normalized column headers, preventing cross-row attribute conflation in online table-to-graph conversion without LLM extraction overhead.
- *Bounded two-tier execution protocol*: An adaptive inference design that concludes unambiguous queries in a single primary call while reserving a targeted second call for diagnostic-resolved states via a minimal sufficient state capsule $\mathcal{K}_2$.

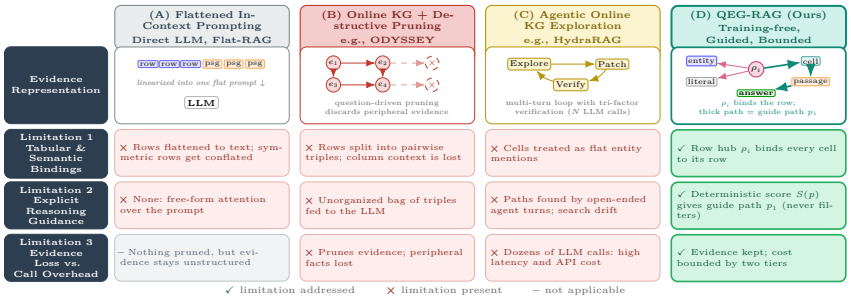


Fig. 1. Architectural comparison across four query-time table–text reasoning paradigms. This comparison strictly evaluates query-time evidence construction and inference over ad-hoc inputs, excluding offline corpus-wide pre-indexing (e.g., GraphRAG, HippoRAG 2) and parametric model training or fine-tuning (e.g., HYBRIDER, MuGER²). In contrast to unguided linear prompt concatenation (a), destructive question-driven graph pruning in online KGs like ODYSSEY (b), and expensive multi-turn agentic exploration over online KGs in systems like HydraRAG and ToG-3 (c), the proposed QEG-RAG framework (d) dynamically instantiates a schema-preserving online multi-graph on-the-fly, retains complete evidence context, guides attention via deterministic multi-factor scored paths, and resolves bottlenecks via a bounded two-tier diagnostic-driven execution protocol.

2 Related Work

Hybrid Table–Text Question Answering and Graph Modeling. Multi-hop reasoning over heterogeneous tabular and textual knowledge was catalyzed by HybridQA [2] and open table–text benchmarks (OTT-QA). Early architectures tackled this challenge via supervised fine-tuning, combining coarse row retrieval with cross-modal neural readers (HYBRIDER [2]), multi-view attention mechanisms (MATE [6]), or document-hopping retrieval (DocHopper [7]). To model complex cross-modal connections, specialized graph frameworks constructed evidence graphs over rows, cells, and sentences (MuGER² [8], S3HQA [9]) trained via graph neural networks (GNNs). While effective, these early systems fundamentally rely on task-specific parameter training over thousands of labeled examples, lacking zero-shot adaptability to ad-hoc tables and producing black-box span extractions without auditable provenance chains. In real-world enterprise RAG, systems must perform strict context-grounded reasoning over unseen schemas without costly re-training, synthesizing answers exclusively from provided evidence without relying on latent parametric memory.

With the advent of instruction-tuned large language models, recent research has shifted toward zero-shot hybrid graph modeling. Most notably, ODYSSEY [3] showed that an online entity graph can be constructed directly at query time from retrieved table cells and passages. However, ODYSSEY highlights two key limitations of existing online graphs: (1) *Loss of Tabular Semantics*: its relation extraction decomposes rows into isolated pairwise entity triples, discarding hierarchical column predicates and causing semantic dilution across symmetric rows; and (2) *Destructive Pruning without Guidance*: to fit LLM context limits, it prunes graph elements based on question token similarity (Fig. 1b). Under strict context-grounding constraints, this aggressive filtering discards periph-

eral supporting evidence and leaves the remaining triples unorganized, forcing the model into ungrounded guessing or empty outputs when candidates tie. In contrast, QEG-RAG preserves the complete retrieved evidence via compact serialization, uses star-topology centroids to maintain row-column bindings, and extracts explicit reasoning path scaffolds to guide LLM attention.

Table Reasoning and Programmatic Decomposition with LLMs. As surveyed by Zhou et al. [1], language models have demonstrated strong tabular capabilities through symbolic code generation or task decomposition. Programmatic frameworks bind LLMs to SQL or Python interpreters (Binder [10], TabSQLify), iteratively decompose queries into sub-questions (DATER [11]), or progressively evolve table structures (Chain-of-Table [12]). Concurrently, TableRAG architectures [13, 14] scale retrieval across large schemas. While programmatic execution excels at structured arithmetic over isolated clean tables, it falters in hybrid multi-hop settings where the critical reasoning step requires traversing unstructured text passages linked to table cells. Code interpreters cannot natively resolve fuzzy semantic transitions across free text without fragile ad-hoc prompts or multi-turn sandbox loops. QEG-RAG bridges this gap by unifying semi-structured table cells and external text passages into a single symbolic evidence multigraph evaluated in a bounded, code-free execution pipeline.

Offline Knowledge Graphs vs. Query-Time Online Knowledge Graphs (Online KGs). Augmenting language models with structured knowledge graphs captures non-local multi-hop dependencies beyond surface retrieval, as demonstrated by QA-GNN [15], JointLK [16], and G-Retriever [17]. In retrieval-augmented generation, persistent frameworks like GraphRAG [18], HippoRAG 2 [19], and KG-CQR [20] pre-construct offline global entity graphs, personalized PageRank memory structures, or corpus-centric relation representations over text corpora before query time. While effective for global text summarization and associative recall, offline GraphRAG incurs prohibitive pre-indexing token overhead, cannot ingest dynamic, ad-hoc tables at query time without re-indexing, and lacks fine-grained row-cell relational bindings.

In contrast to offline indexing, recent frameworks dynamically construct online knowledge graphs (Online KGs) on-the-fly: HydraRAG [5] instantiates an online KG across heterogeneous sources and explores it via tri-factor verification cycles, while ToG-3 [21] deploys multi-agent collaborative loops (Constructor, Retriever, Reflector) over dynamic graphs. However, these agentic online KG paradigms rely on unbounded multi-turn LLM loops (Fig. 1c), resulting in high inference latency and dozens of LLM calls per query. Furthermore, their relation extraction treats table cells as flat entity mentions without record centroid bindings (ρ_i), leaving them vulnerable to cross-row attribute conflation. Crucially, the architectural comparison in Fig. 1 strictly evaluates such query-time execution over ad-hoc inputs, excluding offline corpus-wide pre-indexing and task-specific model training. QEG-RAG addresses both paradigms: unlike offline GraphRAG, it operates completely online per query without pre-indexing; and unlike unbounded agentic online KGs (HydraRAG, ToG-3), QEG-RAG builds a

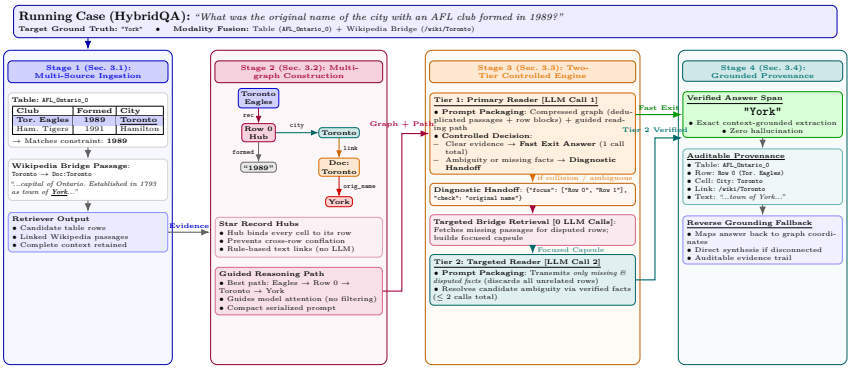


Fig. 2. End-to-end architecture of QEG-RAG. (Sec. 3.1) Multi-Source Evidence Ingestion retrieves relevant table rows and linked passages. (Sec. 3.2) Schema-Preserving Multigraph Construction builds star record hubs and text relations without LLMs. (Sec. 3.3) Two-Tier Controlled Reasoning Engine: Tier 1 reads the compressed graph guided by the top reasoning path; clear cases exit immediately in one call, while ambiguous collisions trigger a diagnostic handoff and targeted bridge retrieval for conclusive resolution in Tier 2. (Sec. 3.4) Grounded Generation and Provenance Audit outputs verified spans with full attribution traces.

schema-preserving multigraph deterministically (0 LLM calls), scores reasoning paths deterministically, and bounds execution to 1 primary call (with at most 1 targeted second call if ambiguous).

3 The QEG-RAG Framework

To systematically resolve these challenges, QEG-RAG orchestrates four cohesive stages: Stage 1 ingests multimodal evidence; Stage 2 constructs an LLM-free schema-preserving multigraph enforcing tabular semantics (ρ_i); Stage 3 scaffolds reasoning paths ($S(p)$) and executes diagnostic-driven retrieval feedback over a bounded two-tier execution protocol ($\mathcal{K}_2$); and Stage 4 conducts grounded generation with provenance auditing.

Problem Formulation and Graph Representation. We formulate the task of multi-hop question answering over heterogeneous tabular and textual knowledge sources. An input instance is defined as a tuple $\mathcal{I} = (q, T, \mathcal{D}_T)$, where q is a natural language question; $T = (C, R)$ is a semi-structured table comprising ordered headers $C = (c_1, \ldots, c_m)$ and rows $R = \{r_1, \ldots, r_n\}$ with discrete cells $v_{i,j}$; and $\mathcal{D}_T = \{d_1, \ldots, d_K\}$ is the corpus of external text passages associated with T, where cell entities connect to external passages via relational mappings $\ell(v_{i,j}) \to d_k$. The objective is to extract the grounded answer span $a^* \in \mathcal{A}$ along with an auditable reasoning path p^*, where a^* may reside as a table cell ($a^* = v_{i,j}$) or as an unstructured text span within a linked passage ($a^* \subset d_k$).

Online Evidence Multigraph Representation. Unlike persistent knowledge base settings that depend on static global entity graphs, table–text RAG requires

synthesizing unstructured text with tabular structures on-the-fly. For each query q, we dynamically instantiate a bounded, query-specific directed multigraph $G_q = (V_q, E_q, \psi_V, \psi_E)$ whose node and edge sets are partitioned as:

$$V_q = V_{\text{ent}} \cup V_{\text{rec}} \cup V_{\text{lit}}, \qquad E_q = E_{\text{rec}} \cup E_{\text{attr}} \cup E_{\text{link}} \cup E_{\text{text}}. \tag{1}$$

The node partition divides vertices by semantic abstraction:

- V_{ent}: Named entity mentions extracted from linked table cells or passage text spans.
- $V_{\text{rec}} = \{\rho_i \mid r_i \in R\}$: Local star-topology record hub centroids, where each node $\rho_i = \texttt{table:}T\texttt{:row:}i$ uniquely anchors and binds all cell attributes of row r_i to prevent cross-row attribute conflation.
- V_{lit}: Unlinked literal values (e.g., numbers, scores, dates, unlinked strings).

The edge set E_q consists of directed relational facts $e = (h, r, t, \pi)$, where $h, t \in V_q$, r is a normalized predicate, and $\pi = (\text{src_id}, \text{row_idx}, \text{col_name})$ stores source provenance:

- $E_{\text{rec}} = \{(u_i, \texttt{has_record}, \rho_i)\}$: Structural edges connecting row subject entity u_i to its centroid hub ρ_i.
- $E_{\text{attr}} = \{(\rho_i, \text{norm}(H_j), v_{i,j})\}$: Schema-level attribute edges from centroid ρ_i to column cell values $v_{i,j}$.
- $E_{\text{link}} = \{(v_{i,j}, \texttt{linked_passage}, d_k)\}$: Cross-modal linking edges connecting candidate cells to associated external text passages.
- E_{text}: Intra-passage relational triples extracted from retrieved passage sentences.

Schema-Preservation and Non-Conflation Property. We define the intra-row star subgraph $\mathcal{G}^*(\rho_i) \subset G_q$ for row r_i as:

$$\begin{aligned}\mathcal{G}^*(\rho_i) = \Big(&\{\rho_i, u_i\} \cup \{v_{i,j}\}_{j=1}^{m}, \\ &\{(u_i, \texttt{has_record}, \rho_i)\} \cup \{(\rho_i, \text{norm}(H_j), v_{i,j})\}_{j=1}^{m}\Big).\end{aligned} \tag{2}$$

By construction, for any distinct table rows $r_i, r_{i'} \in R$ ($i \neq i'$), their star centroids are strictly disjoint: $V(\mathcal{G}^*(\rho_i)) \cap V(\mathcal{G}^*(\rho_{i'})) \cap V_{\text{rec}} = \emptyset$. Consequently, every cell value $v_{i,j}$ maintains a unique 1-hop path to its parent record centroid ρ_i, structurally preventing horizontal cross-row attribute conflation across symmetric candidate rows without requiring destructive graph pruning.

Once question answering concludes, the dynamic multigraph G_q is discarded, incurring zero persistent graph database index overhead. Throughout this section, we illustrate QEG-RAG via the running HybridQA example depicted in Fig. 2: *"What was the original name of the city with an AFL club formed in 1989?"* (ground truth: `York`).

3.1 Stage 1: Multi-Source Evidence Ingestion

Given an input table and an associated corpus of external text passages, retrieval operates dynamically at query time:

- *Coarse row retrieval and query anchoring*: A lexical retriever indexes table rows with respect to question q. A deterministic query anchor inspects q for ordinal constraints (e.g., "second", "last") and exact schema constraints (column header mentions, numerical filters), selecting top-k_1 rows. In Fig. 2, the constraint "formed in 1989" anchors Row 0 (`Tor. Eagles`).
- *Passage expansion via cross-modal linking*: Rather than retrieving entire document corpora indiscriminately, the passage expander resolves relevant external text passages linked to candidate entities in the top-k_1 retrieved rows (via hyperlinks or entity linking) up to a budget of k_2 passages. In Fig. 2, the bridge cell `Toronto` in Row 0 links to `Doc:Toronto`.

3.2 Stage 2: Schema-Preserving Multigraph Construction

Using the retrieved rows and passages, the system dynamically instantiates the multigraph $G_q = (V_q, E_q)$ without LLM calls:

- *Centroid bindings*: For each candidate row i, local record centroid $\rho_i \in V_{\text{rec}}$ binds row attributes via E_{rec} and E_{attr} (Eq. 2).
- *Cross-modal bridging and text relation extraction*: Edges E_{link} connect table cell entities to external passage nodes (e.g., Toronto → Doc:Toronto). Rule-based text triples E_{text} link entities to textual attributes (e.g., Doc:Toronto → York) deterministically without LLM calls.

3.3 Stage 3: Two-Tier Controlled Reasoning Engine

QEG-RAG resolves the operational dilemma between destructive graph pruning and unbounded agentic loops. As illustrated in Fig. 3, the reasoning engine coordinates four transparent steps across two strictly bounded execution tiers:

Tier 1: Primary Reader and Prompt Packaging (First LLM Call). Tier 1 evaluates the initial multimodal evidence without destructive pruning. To supply complete evidence within context limits, we formulate a compressed prompt packaging protocol:

$$\mathcal{X}^*(G_q) = \text{Enc}_{\text{table}}\left(\{\mathcal{G}^*(\rho_i)\}_{i=1}^{k_1}\right) \;\|\; \text{Enc}_{\text{text}}(\mathcal{D}_{\text{unique}}), \tag{3}$$

where table rows are serialized into structured centroid blocks preserving column headers, while external passages are deduplicated and indexed once by source identifier in $\mathcal{D}_{\text{unique}}$. This prompt compression eliminates textual redundancy while retaining complete retrieved evidence facts. Alongside this unpruned context, candidate reasoning paths $p = (v_0, e_1, \ldots, e_L, v_L)$ are scored via a multi-factor linear objective:

$$S(p) = \mathbf{w}^\top \boldsymbol{\phi}(p, q) - \lambda \cdot |p|, \tag{4}$$

where $\boldsymbol{\phi}(p, q)$ captures question entity overlap, table–text transitions, row consistency, and entity support, while $\lambda \cdot |p|$ penalizes redundant hops. The top-scoring

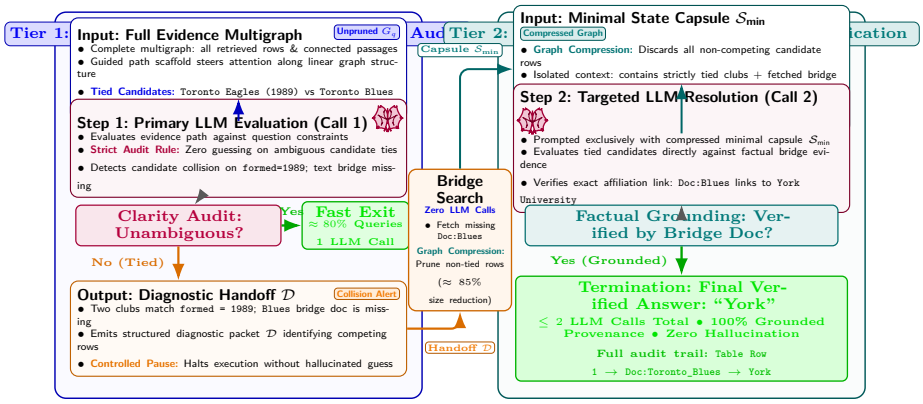


Fig. 3. Two-tier controlled reasoning engine (Stage 3) structured as two parallel algorithmic blocks with deterministic graph compression, illustrated via the running AFL example. (Left) Tier 1 evaluates the unpruned evidence graph under guided path scaffolding. When candidate evidence is unambiguous, execution terminates via Fast Exit in a single LLM call (≈ 80% of queries). When candidate collision occurs (e.g., both Toronto Eagles and Toronto Blues formed in 1989), the LLM emits a structured diagnostic packet. (Center to Right) An automated zero-LLM search fetches targeted bridge passages and executes algorithmic graph compression (≈ 85% size reduction), discarding non-competing candidates to construct the minimal state capsule. In Tier 2, the LLM resolves the ambiguity deterministically by factual evidence in a targeted second call (at most 2 calls total), producing the verified answer with full provenance.

path $p_1 = \arg\max_p S(p)$ is injected into the prompt as a non-filtering reading scaffold steering model attention along verified cross-modal relations.

The language model $\mathcal{M}$ is then invoked in its primary call over prompt $\mathcal{X}_1 = (q, \mathcal{X}^*(G_q), p_1)$. Let $\mathcal{C}_q$ denote the set of candidate record centroids supported by top-scoring paths. The model executes an adaptive audit decision:

- *Fast Exit (Determined State)*: If candidate evidence is unambiguous ($|\mathcal{C}_q| = 1$) and the reasoning path is intact, the model outputs the verified answer span directly, $a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{X}_1)$, concluding execution in a single LLM call (≈ 80% of queries).
- *Diagnostic Handoff (Under-Determined State)*: If candidate collisions occur across symmetric rows ($|\mathcal{C}_q| > 1$) or a bridge link is missing, the model is strictly prevented from guessing. Instead, it emits a structured diagnostic packet $\mathcal{H}_{\text{diag}} = (\mathcal{C}_{\text{focus}}, \beta, \psi_{\text{check}})$, specifying the competing candidate rows $\mathcal{C}_{\text{focus}}$, blocker category β, and the unresolved factual predicate ψ_{check}.

Diagnostic Handoff and Targeted Bridge Augmentation. Upon receiving $\mathcal{H}_{\text{diag}}$, an automated retriever (0 LLM calls) fetches missing candidate bridge passages $\Delta\mathcal{D}$ exclusively for the disputed candidate rows $\mathcal{C}_{\text{focus}}$:

$$\Delta\mathcal{D} = \bigcup_{c \in \mathcal{C}_{\text{focus}}} \left(\ell(c) \setminus \mathcal{D}_{\text{current}}\right), \tag{5}$$

enforcing a symmetric per-candidate retrieval quota to maintain evidentiary balance.

Tier 2: Targeted Reader and State Capsule Packaging (Second LLM Call). Crucially, rather than re-transmitting the broad multigraph, the prompt for Tier 2 packages *exclusively the missing and disputed evidence* into a minimal sufficient state capsule $\mathcal{K}_2$:

$$\mathcal{K}_2 = (q,\, \psi_{\text{check}},\, \{\mathcal{G}^*(\rho_c)\}_{c \in \mathcal{C}_{\text{focus}}},\, \Delta\mathcal{D})\,. \tag{6}$$

By filtering out all unrelated candidate rows and retaining only the disputed record centroids and newly fetched bridge passages $\Delta\mathcal{D}$, $\mathcal{K}_2$ isolates the exact factual differentiator. The language model $\mathcal{M}$ is then invoked in a targeted second call to perform conclusive resolution:

$$a^* = \arg\max_{a \in \mathcal{A}} P_{\mathcal{M}}(a \mid \mathcal{K}_2). \tag{7}$$

This targeted packaging prevents distractor noise and guarantees factual disambiguation without heuristic tiebreaking, with total execution bounded by at most two LLM calls.

3.4 Stage 4: Grounded Generation and Provenance Audit

Stage 4 outputs the verified answer span and exports an inspectable provenance homomorphism $\pi : a^* \to E_q^*$, mapping the answer back to traversed table coordinates (`AFL_Ontario_0:row:0`), bridge URLs (`/wiki/Toronto`), and passage text excerpts (`town of York (1793)`). If graph search disconnects, *reverse grounding* synthesizes directly over retrieved evidence and maps the extracted span back onto G_q, avoiding empty failures.

4 Experimental Protocol

4.1 Benchmark Datasets and Evaluation Metrics

We evaluate across two challenging multi-hop table–text benchmarks ($N = 1{,}500$ each): (1) *HybridQA* [2], evaluating multi-hop reasoning over Wikipedia tables aligned with hyperlinked passages; and (2) *OTT-QA* [22], evaluating open table-and-text retrieval and question answering where candidate tables and text spans must be fetched and grounded jointly. Both benchmark subsets strictly preserve the stratified heterogeneous reasoning distribution across numeric and textual answer modalities.

Crucially, both benchmarks feature a substantial proportion of inherently ambiguous queries: multiple symmetric table rows frequently satisfy initial filtering constraints (e.g., identical years, categories, or scores), creating severe candidate collisions ($|\mathcal{C}_q| > 1$), while external passage hyperlinks may be missing, incomplete, or require multi-hop bridge traversal. Under strict context grounding, these ambiguous cases easily induce hallucinated guesses in unguided LLMs or catastrophic over-abstention in heuristically pruned graph methods, making them an ideal testbed for evaluating diagnostic disambiguation.

Evaluation Scope: Zero-Shot In-Context RAG vs. Supervised Fine-Tuning. We explicitly clarify the operational scope of this evaluation. While task-specific supervised models (e.g., MuGER[2] [8], S3HQA [9]) achieve higher leaderboard numbers by fine-tuning millions of parameters over the 62k labeled training instances of HybridQA, they are specialized to fixed benchmark distributions and cannot generalize to ad-hoc, out-of-domain tables without expensive re-annotation and re-training. In contrast, QEG-RAG operates in the strictly training-free, zero-shot in-context RAG regime (0 training examples, 0 updated weights). Under this practical setting, our primary comparisons are appropriately framed against published zero-shot hybrid graph frameworks (e.g., ODYSSEY [3]) and in-context LLM baselines under identical context and retrieval budgets.

Strict Context-Grounded Protocol. To rigorously assess faithful reasoning and prevent models from relying on latent parametric memorization (e.g., pre-trained Wikipedia facts), all systems are evaluated under a strict context-grounding protocol. Prompts mandate that answers must be derived exclusively from the provided tabular and textual evidence; when the factual chain is broken or evidence is insufficient, the model must output an ungrounded or empty response rather than guessing. This protocol faithfully mirrors enterprise and scientific environments where hallucination is unacceptable.

We report three standard accuracy metrics: (1) *Exact Match (EM)* under standard normalization; (2) *Token-level F1*; and (3) *Semantic Exact Match (Semantic EM)* tolerating surface aliases. System efficiency is reported via average LLM calls and ungrounded empty answer percentage.

4.2 Baselines and Experimental Settings

Model Paradigms and Benchmark Scope. All methods are evaluated under strictly standardized retrieval pools ($k_1 = 5, k_2 = 3$), context budgets (24,000 characters), and greedy decoding ($T = 0$). We evaluate across two distinct model paradigms:

- *Primary Benchmark Suite (Native Reasoning Models)*: Our core benchmark matrix focuses on modern models with native test-time reasoning tokens: GPT-OSS 20B and GPT-OSS 120B. These models possess internal chain-of-thought capabilities and are evaluated comprehensively across both HybridQA and OTT-QA ($N = 1{,}500$ each).
- *Cross-Paradigm Generalization Study (Standard Non-Reasoning LLMs)*: To examine generalization beyond native reasoning architectures, we evaluate Qwen 122B (`Qwen2.5-72B/122B-Instruct`), a flagship instruction-tuned LLM without test-time reasoning tokens. *We emphasize that Qwen 122B is not part of the primary benchmark suite*; rather, it serves as a focused investigation on OTT-QA ($N = 1{,}500$) into whether QEG-RAG's neuro-symbolic scaffolding can independently rescue non-reasoning models from the severe over-abstention trap induced by strict grounding constraints.

Compared Configurations. We compare five representative configurations under identical context budgets: (1) *Direct LLM*: Standard RAG prompting over

Table 1. Comprehensive comparative evaluation of native reasoning models across multi-hop table–text benchmarks under strict context grounding ($k_1 = 5, k_2 = 3$, 24k context budget, $N = 1{,}500$ per benchmark).

Dataset	Model	Method	EM (%)	F_1 (%)	Sem. EM	Empty	Calls
HybridQA	GPT-OSS 20B	Direct LLM	45.73	54.25	48.00	30.20%	**1.00**
		ODYSSEY [3]	32.80	39.98	34.80	41.87%	5.39
		Graph w/o Path	31.80	37.92	32.93	46.00%	**1.00**
		Path-Text Top-N	10.87	12.53	10.93	79.13%	**1.00**
		QEG-RAG (Ours)	**53.07**	**66.45**	**54.07**	**11.53%**	1.23
	GPT-OSS 120B	Direct LLM	56.20	65.62	57.87	20.00%	**1.00**
		ODYSSEY [3]	37.80	45.17	39.13	38.87%	5.38
		Graph w/o Path	33.93	39.80	34.67	48.07%	**1.00**
		Path-Text Top-N	13.07	15.13	13.13	75.53%	**1.00**
		QEG-RAG (Ours)	**59.27**	**70.70**	**60.00**	**10.73%**	1.20
OTT-QA	GPT-OSS 20B	Direct LLM	50.80	57.70	51.80	30.00%	**1.00**
		ODYSSEY [3]	37.47	43.66	38.60	40.73%	5.44
		Graph w/o Path	40.87	46.35	41.60	39.53%	**1.00**
		Path-Text Top-N	9.40	10.39	9.47	82.20%	**1.00**
		QEG-RAG (Ours)	**55.60**	**67.59**	**55.93**	**11.87%**	1.24
	GPT-OSS 120B	Direct LLM	60.40	67.93	61.20	21.20%	**1.00**
		ODYSSEY [3]	41.73	47.96	42.53	39.47%	5.48
		Graph w/o Path	42.27	47.58	42.87	40.87%	**1.00**
		Path-Text Top-N	13.20	14.88	13.27	75.93%	**1.00**
		QEG-RAG (Ours)	**62.93**	**73.03**	**63.07**	**10.80%**	1.22

flattened table rows and text passages; (2) *ODYSSEY* [3]: A zero-shot online hybrid graph with aggressive question-token pruning; (3) *Graph w/o Path*: Our complete, unpruned schema-preserving multigraph supplied without linear path sequences; (4) *Path-Text Top-N*: Hard context filtering retaining exclusively the top-N ranked traversal paths; and (5) *QEG-RAG (Ours)*: The complete framework integrating lossless prompt compression, deterministic path scaffolding, and the two-tier diagnostic reasoning engine.

5 Results and In-Depth Analysis

To systematically assess the empirical performance and architectural mechanisms of QEG-RAG, we formulate and address five key research questions grounded in quantitative benchmark evidence:

RQ1 (Main Benchmark Grounding Performance): Under strict context grounding and prevalent retrieval ambiguities, does QEG-RAG achieve superior grounding accuracy and mitigate ungrounded failures across heterogeneous benchmarks compared to direct prompting and hybrid graph baselines?

Empirical Findings: Table 1 reports primary results across HybridQA and OTT-QA ($N = 1{,}500$ each) on native reasoning models. QEG-RAG consistently achieves the highest grounding accuracy. On HybridQA with GPT-OSS 20B, QEG-RAG reaches 53.07% EM and 66.45% F_1 (+7.34% EM over Direct LLM; +20.27% over ODYSSEY); on 120B, it achieves 59.27% EM and 70.70% F_1 (+3.07% EM over Direct LLM; +21.47% over ODYSSEY, McNemar's $p < 0.001$). On open retrieval (OTT-QA), where candidate tables and text are retrieved jointly—greatly amplifying candidate collisions—QEG-RAG demonstrates identical gains, reaching 55.60% EM on 20B (+4.80%) and 62.93% EM on 120B (+2.53%).

In-Depth Diagnostic Analysis: A closer inspection reveals why baselines collapse on ambiguous multi-hop cases under strict context grounding: (1) *Direct LLM*: Linearizing tables strips schema boundaries. When ambiguous queries match multiple symmetric rows satisfying partial constraints (e.g., several teams formed in 1989), attention disperses, blurring row-column bindings and causing cross-row conflation or ungrounded guessing (yielding 20.00%–30.20% empty

Table 2. Cross-paradigm generalization on standard non-reasoning instruction-tuned LLM (Qwen 122B on OTT-QA, $N = 1{,}500$). QEG-RAG acts as an external neuro-symbolic scaffold, eliminating empty abstentions and elevating EM.

Method	EM (%)	F_1 (%)	Sem. EM	Empty	Calls
Direct LLM	29.27	31.52	29.53	65.80%	**1.00**
ODYSSEY [3]	5.93	6.99	5.93	92.07%	6.66
Graph w/o Path	9.93	11.00	10.00	88.47%	**1.00**
Path-Text Top-N	1.67	1.89	1.73	97.53%	**1.00**
QEG-RAG (Ours)	**33.40**	**35.30**	**33.53**	**0.00%**	1.64

responses). (2) *ODYSSEY & Path-Text*: Decomposing tables into pairwise triples with aggressive question-token pruning severs multi-hop chains whenever resolving bridge facts reside in peripheral cells or external passages lacking lexical overlap with the query. Consequently, ODYSSEY incurs 38.87%–41.87% empty abstentions (EM 32.80%–41.73%), while hard path filtering (Path-Text Top-N) collapses to 9.40%–13.20% EM (75.53%–82.20% empty). In contrast, QEG-RAG avoids pruning via lossless multigraph compression, steers attention along verified relations with path scaffolding (p_1), and resolves ambiguous collisions ($|\mathcal{C}_q| > 1$) via targeted symmetric bridge retrieval in Tier 2. This suppresses empty failures to 10.73%–11.87% (2.5–4× below ODYSSEY) while maximizing faithful exact match.

RQ2 (Cross-Paradigm Generalization to Non-Reasoning LLMs): Can QEG-RAG generalize beyond native reasoning architectures and rescue standard instruction-tuned LLMs from the strict-grounding over-abstention trap?

Finding: Outside the main benchmark, Table 2 evaluates cross-paradigm generalization on Qwen 122B (OTT-QA). Under strict grounding without test-time reasoning tokens, standard instruction-tuned LLMs fall into a catastrophic *over-abstention trap*: Direct LLM emits 65.80% empty responses (29.27% EM), while ODYSSEY collapses to 92.07% empty responses (5.93% EM) because fragmented pairwise triples fail to provide cohesive grounding. In sharp contrast, QEG-RAG functions as an external neuro-symbolic reasoning scaffold, completely eliminating empty failures (0.00% Empty) and elevating EM to 33.40% (+4.13% over Direct LLM; 5.6× higher than ODYSSEY's 5.93%). This validates that structured evidence graphs and deterministic path scaffolding can substitute for native internal reasoning tokens.

RQ3 (Explicit Reasoning Guidance vs. Destructive Pruning): How effectively does deterministic path scaffolding guide LLM attention along multi-hop chains, and why does aggressive graph pruning cause catastrophic failure?

Finding: Across Tables 1 and 2, hard path filtering (Path-Text Top-N) collapses to 1.67%–13.20% EM with 75.53%–97.53% empty abstentions because cutting peripheral edges severs multi-hop reasoning chains. In Table 3 (Row 6), supplying the unpruned multigraph without path sequences (Graph w/o Path) still retains 61.60% EM on 120B, proving that QEG's schema-preserving multigraph topology inherently provides the necessary multi-hop reach. However, providing unorganized edge sets disperses LLM attention. Supplying path p_1 directs focus along verified relations, yielding 92.0% chain faithfulness in our manual audit. Furthermore, in Table 3 (Row 1), disabling lossless prompt compression causes EM to collapse by −27.60% with 52.40% empty answers due to context blowout, demonstrating the critical necessity of lossless serialization.

Table 3. Systematic knockout ablation study on pooled partitions ($N = 500$ under GPT-OSS 120B). All variants isolate individual components against full QEG-RAG.

Ablation Configuration	EM (%)	F1 (%)	ΔEM	Empty	Architectural Implication
Full QEG-RAG (Two-Tier)	**59.20**	**68.06**	—	**11.40%**	Full controlled evidence scaffolding
(1) w/o Prompt Compression	31.60	37.36	−27.60	52.40%	Token blowout; > 52% empty abstentions
(2) w/o Record Nodes ρ_i	56.20	62.74	−3.00	19.60%	Cross-row conflation; doubles empty rate
(3) w/o Tier-2 Targeted Evidence	56.00	63.60	−3.20	18.40%	Re-reading misses external bridge facts
(4) w/o Diagnostic Focus	57.80	65.74	−1.40	15.60%	Lack of cell-level grounding harms F1
(5) w/o Adaptive Tier-2 (Single Tier-1)	58.60	66.78	−0.60	5.40%	Single tier fails on collisions (−7.68% F1)
(6) w/o Path Guidance (Graph)	61.60	70.14	+2.40	11.80%	Flexible reach via multigraph topology

RQ4 (Diagnostic Feedback and Bounded Execution Efficiency): Can the two-tier diagnostic protocol resolve retrieval collisions factually while strictly bounding execution overhead against unbounded agent loops?

Finding: Unlike unbounded multi-turn agent loops (e.g., HydraRAG) or heuristic tiebreakers, QEG-RAG concludes unambiguous queries in a single primary call (Fast Exit, $\approx 80\%$ of queries), averaging only 1.20–1.24 calls per query on native reasoning models and 1.64 calls on Qwen 122B (Tables 1 and 2), compared to 5.38–6.66 calls for ODYSSEY. When structural collisions ($|\mathcal{C}_q| > 1$) or missing passages occur, the diagnostic handoff triggers targeted Tier-2 verification. In Table 3, omitting targeted Tier-2 evidence (Row 3) degrades performance by -3.20% EM and elevates empty abstentions to 18.40%; omitting diagnostic focus (Row 4) drops EM by -1.40%; and omitting Tier 2 entirely (Row 5, single Tier-1) drops F1 by -7.68% on ambiguous collisions, confirming the efficiency and efficacy of diagnostic bridging.

RQ5 (Preservation of Tabular Semantics): Does the star centroid formulation prevent cross-row attribute conflation under symmetric candidate rows without LLM extraction costs?

Finding: In Direct LLM, linearizing tables decouples cell values from parent rows, causing attribute conflation when multiple symmetric rows satisfy partial conditions. In Table 3 (Row 2), eliminating star centroids ρ_i drops EM by -3.00% and F1 by -5.32%, while nearly doubling empty abstentions from 11.40% to 19.60%. This confirms that star centroids mathematically enforce Row Attribute Non-Conflation, anchoring cell attributes to column predicates without costly LLM relation extraction.

Summary of Knockout Ablations & Provenance Audit. Table 3 confirms all architectural pillars are non-redundant. A manual audit on 100 questions confirms: (1) 92.0% faithful traces to verified evidence; (2) 89.0% ground-truth bridge alignment; and (3) reduction of empty failures on 120B to 0.53% (vs. 3.40% for ODYSSEY).

6 Limitations

While QEG-RAG effectively structures and reasons over retrieved context without destructive pruning, downstream performance remains bounded by the completeness of upstream evidence retrieval. Further advancing evidence search and multi-hop retrieval strategies presents a direct pathway to further elevate Exact Match (EM) accuracy on complex questions.

7 Conclusion

We presented QEG-RAG, uniting query-time evidence graphs with path-scaffolded diagnostic reasoning for faithful table–text question answering. By combining schema-preserving multigraphs, compact lossless compression, deterministic path scaffolding, and adaptive two-tier diagnostic verification, QEG-RAG resolves the dilemma between destructive graph pruning and unbounded agent loops. Across both reasoning and non-reasoning paradigms under strict context grounding, QEG-RAG consistently achieves higher grounding accuracy, eliminates empty failures, and provides auditable provenance within a bounded 1-to-2 call budget.

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